

Uniform Spanning Tree / LERW Random Environment Project Explainer

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Abstract

This note explains the whole current state of the numerical project on loop-erased random walk (LERW) winding in fixed random environments. It covers the mathematical motivation, the implemented models, the simulation pipeline, the diagnostics used to detect drift, the focused numerical results up to $L = 256$, the supervisor feedback, and the next larger-run plan. The most important current conclusion is that the original two-random-weight model develops strong north-east drift after probability normalisation. The next scientific direction is therefore to use locally balanced random environments, where every lattice site receives its own independent four-direction sample, and then test larger sizes and alternative distributions.

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1 Executive Summary

The project studies random walks on the square lattice box

$$B_L = [-L, L]^2 \cap \mathbb{Z}^2,$$

started at the origin and stopped when the walk first hits the boundary. The chronological loop erasure of the walk gives a loop-erased random walk path. The main observable is the winding,

$$W_L = \#\{\text{left turns}\} - \#\{\text{right turns}\}.$$

For the ordinary simple symmetric random walk, the expected qualitative behaviour is

$$\text{Var}(W_L) \sim C \log L.$$

The random-environment question is whether a fixed site-dependent environment can produce anomalous growth closer to

$$\text{Var}(W_L) \sim C(\log L)^2.$$

The initial implementation tested a two-random-weight edge model

$$w_N = 1, \quad w_E = 1, \quad w_S = u, \quad w_W = v,$$

with u, v positive mean-one random variables. After normalisation into transition probabilities, this model produces a strong effective north-east drift for moderate and strong disorder. That drift makes the walk exit near the north-east corner and suppresses winding variance. It is therefore not a clean test of a centred random-environment winding question.

The latest supervisor feedback confirms the next direction:

- use a more locally balanced random environment;
- ensure each site has a different random sample;
- run larger simulations;
- if the balanced Gamma model does not show clear $(\log L)^2$ behaviour at larger sizes, test other distribution choices.

The code now supports this plan through four-weight locally balanced models: `gamma4`, `lognormal4`, `pareto4`, and `uniform4`.

2 Mathematical Background

2.1 Uniform Spanning Tree and LERW

Wilson’s algorithm samples a uniform spanning tree (UST) by repeatedly running random walks and adding their loop erasures to the tree. Thus branches of the UST are described by loop-erased random walk. Studying the geometry of LERW is therefore a way of studying geometric properties of the UST.

In this project the numerical experiment is not sampling full spanning trees. Instead, it focuses on one branch-like observable: run a random walk from the origin until it exits a square box, erase loops chronologically, and measure the winding of the resulting self-avoiding path.

2.2 Winding Observable

The square lattice directions are encoded cyclically as

$$E = 0, \quad N = 1, \quad W = 2, \quad S = 3.$$

For consecutive LERW steps, if the direction changes by $+1$ modulo 4, this is counted as a left turn. If it changes by -1 modulo 4, this is counted as a right turn. The winding is

$$W_L = L_L - R_L,$$

where L_L is the number of left turns and R_L is the number of right turns along the loop-erased path.

Immediate U-turns should not occur in a loop-erased nearest-neighbour path. The code raises an error if one is encountered, because it would indicate a bug or an invalid path.

2.3 Scaling Question

The core numerical question is whether the winding variance grows like

$$C \log L$$

or whether a fixed random environment can produce a stronger law

$$C(\log L)^2.$$

All numerical fits in this project are descriptive. They are not proofs of asymptotic behaviour. With only finitely many sizes, especially sizes only up to $L = 256$, both $\log L$ and $(\log L)^2$ curves can sometimes look plausible. The diagnostics must therefore be interpreted conservatively.

3 Simulation Setup

3.1 Domain and Stopping Rule

For each box size L , the walk starts at

$$X_0 = (0, 0)$$

and evolves until the first time

$$\tau_L = \inf\{t \geq 0 : X_t \in \partial B_L\},$$

where the boundary condition is implemented as

$$|x| = L \quad \text{or} \quad |y| = L.$$

The raw path is

$$(X_0, X_1, \dots, X_{\tau_L}).$$

The code then chronologically erases loops to obtain the LERW path

$$\text{LE}(X_0, \dots, X_{\tau_L}).$$

3.2 Recorded Trial Data

Each simulated trial records:

- box size L ;
- model name;
- distribution parameter, stored in the CSV column k ;
- sample index;
- random seed;
- raw walk length;
- LERW length;
- winding;
- boundary hit coordinates;
- number of random-environment sites sampled.

The main CSV schema is:

```
L,model,k,sample_index,seed,raw_steps,lerw_steps,winding,  
hit_x,hit_y,environment_sites
```

3.3 Lazy Fixed Environments

The random environment is sampled lazily. When the walk first visits a site, the code samples that site's random weights and stores them in a dictionary. Future visits to the same site reuse the same weights.

This is statistically equivalent to pre-sampling the entire box because the site environments are independent. Laziness only avoids wasting time sampling sites that the walk never visits.

This implementation directly addresses the supervisor's clarification that each site should have a different sample.

4 Implemented Models

4.1 Symmetric Baseline

The calibration model is the ordinary simple symmetric random walk:

$$p_N = p_E = p_S = p_W = \frac{1}{4}.$$

This case is expected to show winding variance consistent with $C \log L$.

4.2 Original Two-Random-Weight Gamma Model

The original two-random-weight model uses

$$w_N = 1, \quad w_E = 1, \quad w_S = u, \quad w_W = v,$$

where

$$u, v \sim \Gamma(k, 1/k).$$

The scale $1/k$ makes the raw weights mean one:

$$\mathbb{E}[u] = \mathbb{E}[v] = 1.$$

Transition probabilities are obtained by normalising:

$$p_d(x) = \frac{w_d(x)}{w_N(x) + w_E(x) + w_S(x) + w_W(x)}.$$

Although the raw random variables have mean one, this model is not locally balanced after normalisation. Since north and east are fixed at weight 1, while south and west can be very small, the walk can acquire a strong effective north-east drift.

4.3 Exponential Two-Weight Model

The `exponential` model has the same two-weight structure as the Gamma model, but uses

$$u, v \sim \text{Exp}(1).$$

This is equivalent to the two-weight Gamma model at $k = 1$, but it is kept as a separate named model for clarity.

4.4 Balanced Gamma Model: `gamma4`

The balanced Gamma model samples all four outgoing weights independently:

$$w_N, w_E, w_S, w_W \stackrel{\text{iid}}{\sim} \Gamma(k, 1/k).$$

The four raw weights are then normalised to probabilities at that site. Because all directions are treated symmetrically, this model is locally balanced in distribution.

4.5 Balanced Lognormal Model: `lognormal4`

The balanced lognormal model samples

$$w_N, w_E, w_S, w_W \stackrel{\text{iid}}{\sim} \text{LogNormal}\left(-\frac{\sigma^2}{2}, \sigma\right).$$

The choice of mean parameter $-\sigma^2/2$ ensures

$$\mathbb{E}[w_d] = 1.$$

The model parameter is σ .

4.6 Balanced Pareto Model: `pareto4`

The balanced Pareto model samples

$$w_d = \frac{\alpha - 1}{\alpha} Y_d, \quad Y_d \sim \text{Pareto}(\alpha),$$

independently for $d \in \{N, E, S, W\}$. The parameter must satisfy

$$\alpha > 1$$

so that the mean is finite. The prefactor makes the raw weights mean one. This model gives a heavier-tailed alternative to Gamma and lognormal weights.

4.7 Balanced Uniform Model: uniform4

The balanced uniform model samples

$$w_N, w_E, w_S, w_W \stackrel{\text{iid}}{\sim} \text{Uniform}(1 - a, 1 + a),$$

with

$$0 < a < 1.$$

This gives a bounded-disorder comparison case.

5 Code Structure

The current project folder contains the following main files.

`lerw_random_environment.py`

Core simulator. Defines the random environment, runs random walks to the boundary, performs loop erasure, computes winding, and writes trial CSVs.

`run_research_batch.py`

Batch runner. Runs many model/size/parameter configurations and supports per-configuration checkpoint files. It now supports explicit configs such as `gamma4:1`, `lognormal4:0.5`, `pareto4:2`, and `uniform4:0.5`.

`analyse_winding_results.py`

Command-line summary script. Reads one or more CSVs, groups by model, parameter, and size, then prints group means, variances, and simple scaling fits.

`make_figures.py`

Generates SVG figures, CSV tables, and a Markdown analysis dossier using only the Python standard library.

`test_lerw_random_environment.py`

Unit tests for loop erasure, winding, probability normalisation, fixed-in-time environments, and distinct site samples.

`README.md`

Practical project guide and current command reference.

6 Existing Data and Outputs

6.1 Main Result CSVs

Important result files include:

- `results/research_batch_main.csv`: broad exploratory sweep.
- `results/research_batch_focused.csv`: focused run used for the main analysis below.
- `results/research_batch_focused_*.csv`: per-configuration checkpoints from the focused run.
- `results/smoke_balanced_distributions.csv`: small smoke test for the new balanced distributions.

6.2 Analysis Output Folders

- `analysis_focused/`: focused figures, tables, and interpretation.
- `analysis_output/`: broader exploratory analysis.
- `analysis_smoke_balanced/`: smoke-test output for the new balanced distributions.

7 Focused Numerical Run

The focused run used:

$$L \in \{16, 32, 64, 128, 256\},$$

with 700 samples per model/size/parameter cell. The models were:

- `symmetric`;
- two-weight `gamma`;
- balanced `gamma4`.

For `gamma` and `gamma4`, the tested parameters were

$$k \in \{20, 1, 0.5\}.$$

The focused run contains 24,500 trials.

8 Focused Results: Symmetric Baseline

The symmetric walk behaves as expected and calibrates the simulation.

L	$\text{Var}(W)$	mean raw steps	mean LERW steps
16	1.894	296.9	40.3
32	2.353	1239.1	96.2
64	3.148	4861.2	231.8
128	3.572	20045.7	559.1
256	4.300	77271.2	1329.5

The descriptive fit against $\log L$ is

$$\text{Var}(W) = 0.870 \log L - 0.565, \quad R^2 = 0.992.$$

The descriptive fit against $(\log L)^2$ is

$$\text{Var}(W) = 0.104(\log L)^2 + 1.158, \quad R^2 = 0.987.$$

The log fit is slightly better, as expected for the baseline.

9 Focused Results: Two-Weight Gamma Model

9.1 Drift Diagnostic

At $L = 256$, the mean boundary hit locations for the two-weight Gamma model are:

k	mean hit x/L	mean hit y/L	mean raw steps
20	0.589	0.591	48969.1
1	0.894	0.889	4731.9
0.5	0.916	0.918	2891.9

This is the central modelling issue. For $k = 1$ and $k = 0.5$, the walk exits very near the north-east corner. The drift is not a small numerical fluctuation. It is large enough to change the qualitative behaviour of the raw walk.

9.2 Winding Variance

k	$\text{Var}(W)$ at $L = 16$	$\text{Var}(W)$ at $L = 256$	R^2 for $\log L$	R^2 for $(\log L)^2$
20	1.973	3.735	0.952	0.908
1	1.755	2.099	0.524	0.587
0.5	1.358	1.548	0.950	0.916

The moderate and strong disorder cases have suppressed winding variance. This is consistent with a drifted-walk regime: the walk exits quickly and does not sample large-scale winding in a centred way.

10 Focused Results: Balanced Gamma Model

10.1 Drift Diagnostic

At $L = 256$, the balanced `gamma4` model remains centred:

k	mean hit x/L	mean hit y/L	mean raw steps
20	0.017	-0.027	79695.7
1	0.028	0.026	96802.5
0.5	0.017	-0.003	149512.6

This supports the interpretation that the four-weight model is a more suitable candidate for a no-drift random-environment test.

10.2 Winding Variance

k	$\text{Var}(W)$ at $L = 16$	$\text{Var}(W)$ at $L = 256$	R^2 for $\log L$	R^2 for $(\log L)^2$
20	2.034	4.104	0.963	0.938
1	1.987	4.001	0.978	0.943
0.5	1.965	4.184	0.976	0.946

Over $L \leq 256$, the balanced Gamma model still looks closer to logarithmic growth than to clear $(\log L)^2$ growth. This does not disprove the conjectured behaviour. It only says the currently tested size range is not enough to show it convincingly.

11 Interpretation of the Drift Problem

The two-weight model uses mean-one raw weights, but mean-one raw weights do not guarantee zero effective drift after normalisation. The key issue is structural: north and east are fixed at weight 1, while south and west are random. When south and west weights are often small, north and east become relatively more likely. This creates a systematic north-east preference.

The boundary-hit diagnostic is therefore essential. If the mean boundary hit location is close to $(0,0)$ after scaling by L , the walk is plausibly centred. If the mean hit is close to (L, L) , the walk is drifted and winding variance should not be interpreted as evidence for the centred random environment problem.

12 Supervisor Feedback and Current Direction

The latest feedback from Prof. Chhita confirms that the project should now use a more locally balanced random environment. The important points are:

- each site should have a different sample;
- larger simulations should be run;
- other distributions should be tested if the Gamma distribution does not show $(\log L)^2$ behaviour at larger sizes;
- the conjectured $(\log L)^2$ behaviour is of community interest, so identifying the right distributional setting is valuable.

The code changes made after this feedback implement exactly this workflow:

- `gamma4` is the locally balanced Gamma model;
- `lognormal4`, `pareto4`, and `uniform4` are additional balanced distribution families;
- the tests verify that probabilities are positive and normalised;
- the tests verify that different sites receive different random samples.

13 Recommended Larger Simulation

The recommended next large run is:

```
python3 run_research_batch.py \
--sizes 16 32 64 128 256 512 \
--samples 1000 \
--seed 20260615 \
--configs symmetric gamma4:20 gamma4:1 gamma4:0.5 \
          lognormal4:0.5 lognormal4:1 \
          pareto4:3 pareto4:2 uniform4:0.5 \
--checkpoint-per-config \
--output results/research_batch_balanced_large.csv
```

The `--checkpoint-per-config` option is important. It writes a separate CSV after each model/parameter configuration finishes, so partial results are not lost if a long run is interrupted.

For a quick smoke test before the long run:

```
python3 run_research_batch.py \
  --sizes 16 32 \
  --samples 5 \
  --seed 20260615 \
  --configs symmetric gamma4:1 lognormal4:0.5 pareto4:2 uniform4:0.5 \
  --output results/smoke_balanced_distributions.csv
```

14 How to Regenerate Figures and Tables

For the focused old analysis:

```
python3 make_figures.py \
  results/research_batch_focused.csv \
  --output-dir analysis_focused
```

For the new balanced large run, after it finishes:

```
python3 make_figures.py \
  results/research_batch_balanced_large.csv \
  --output-dir analysis_balanced_large
```

The script writes:

- analysis_*/figures/*.svg;
- analysis_*/tables/group_stats.csv;
- analysis_*/tables/scaling_fits.csv;
- analysis_*/analysis_dossier.md.

15 Figure Guide

The most important generated figures are:

variance_vs_logL_selected.svg

Winding variance against $\log L$ for selected models.

variance_vs_logL2_selected.svg

Same data plotted against $(\log L)^2$.

mean_hit_x_over_L.svg

Drift diagnostic for the horizontal boundary hit coordinate.

mean_hit_y_over_L.svg

Drift diagnostic for the vertical boundary hit coordinate.

boundary_hits_L128.svg

Scatter plot of boundary hit locations at $L = 128$ in the focused analysis.

raw_walk_length.svg

Mean raw walk length. Drifted models exit much faster.

lerw_path_length.svg

Mean loop-erased path length.

environment_vector_fields.svg

Local mean-step vector field visualisation.

16 Testing and Verification

The unit test command is:

```
python3 -m unittest -v
```

The current test suite checks:

- chronological loop erasure;
- winding sign convention;
- symmetric transition probabilities;
- positive and normalised probabilities for Gamma models;
- positive and normalised probabilities for lognormal, Pareto, and uniform balanced models;
- fixed-in-time environments at a site;
- different samples at different sites;
- small boundary-hitting trials.

At the time this explainer was written, the test suite passed:

10/10 tests passing.

17 What Can Be Claimed Safely

The following claims are currently defensible:

- The symmetric baseline reproduces the expected qualitative logarithmic growth.
- The original two-weight Gamma model develops strong effective north-east drift after normalisation.
- Strong drift correlates with shorter raw walks and suppressed winding variance.
- The balanced four-weight Gamma model stays centred in the tested size range.
- Up to $L = 256$, the balanced Gamma model does not show convincing $(\log L)^2$ growth.
- Larger sizes and alternative balanced distributions are the correct next numerical step.

The following claims should be avoided:

- Do not claim the conjectured $(\log L)^2$ behaviour is false.
- Do not claim the random-environment problem is solved.
- Do not call the two-weight model drift-free just because the raw weights have mean one.
- Do not treat one balanced distribution as decisive without checking larger sizes and other distribution families.

18 Potential Paper Structure

A future write-up could be structured as follows:

1. Introduction: UST, LERW, winding, and motivation for random environments.
2. Model definitions: symmetric walk, two-weight model, balanced four-weight models.
3. Numerical method: box, stopping rule, loop erasure, winding convention, seeding, sample sizes.
4. Baseline validation: symmetric $C \log L$ behaviour.
5. Drift diagnostic: two-weight Gamma model and north-east exit bias.
6. Balanced environments: Gamma and alternative distributions.
7. Scaling diagnostics: fits to $\log L$ and $(\log L)^2$.
8. Discussion: finite-size limitations, drift versus disorder, and future larger runs.
9. Conclusion: what is established and what remains open.

19 Open Questions

The main open scientific questions are:

- Does any locally balanced fixed random environment show convincing $(\log L)^2$ winding variance at larger L ?
- If Gamma weights do not show the effect, do heavier-tailed models such as `pareto4` produce it?
- How large must L be before the asymptotic regime becomes visible?
- How many samples are needed for stable variance estimates at $L = 512$ or larger?
- Should future diagnostics include bootstrap confidence intervals or error bars for the variance estimates?

20 Immediate Next Steps

1. Run the documented smoke test if needed.
2. Launch the larger balanced batch with checkpointing.
3. Regenerate figures and tables for the new large CSV.
4. Check boundary-hit diagnostics first. Any model with strong drift should be treated as a drifted model, not a centred random environment.
5. Compare variance fits against $\log L$ and $(\log L)^2$.
6. Prioritise models that are both centred and show possible upward curvature relative to $\log L$.
7. Add error bars before making stronger claims.

A Command Reference

A.1 Run Tests

```
python3 -m unittest -v
```

A.2 Single-Model Debug Run

```
python3 lerw_random_environment.py \  
  --sizes 16 32 64 \  
  --samples 50 \  
  --model gamma4 \  
  --k 1 \  
  --output results/debug_gamma4_k1.csv
```

A.3 Previous Focused Batch

```
python3 run_research_batch.py \  
  --sizes 16 32 64 128 256 \  
  --samples 700 \  
  --seed 20260613 \  
  --models symmetric gamma gamma4 \  
  --k-values 20 1 0.5 \  
  --checkpoint-per-config \  
  --output results/research_batch_focused.csv
```

A.4 New Balanced Batch

```
python3 run_research_batch.py \  
  --sizes 16 32 64 128 256 512 \  
  --samples 1000 \  
  --seed 20260615 \  
  --configs symmetric gamma4:20 gamma4:1 gamma4:0.5 \  
    lognormal4:0.5 lognormal4:1 \  
    pareto4:3 pareto4:2 uniform4:0.5 \  
  --checkpoint-per-config \  
  --output results/research_batch_balanced_large.csv
```

A.5 Analyse a CSV in the Terminal

```
python3 analyse_winding_results.py results/research_batch_focused.csv
```

A.6 Generate Figures

```
python3 make_figures.py \  
  results/research_batch_focused.csv \  
  --output-dir analysis_focused
```

B Glossary

LERW	Loop-erased random walk. It is obtained from a random walk by erasing loops in chronological order.
UST	Uniform spanning tree. Wilson's algorithm constructs it using LERW.

Fixed environment

Random transition probabilities are sampled once per site and then kept fixed throughout the walk.

Locally balanced

All four outgoing directions at a site are sampled symmetrically from the same distribution, so there is no built-in preferred direction in law.

Drift diagnostic

A check based on the mean boundary hit location. Values of mean hit x/L and y/L close to zero indicate a centred walk. Values close to one indicate north-east drift.

Disorder strength

Informal term for how variable the random weights are. Smaller Gamma k , larger lognormal σ , and smaller Pareto α generally correspond to stronger disorder.